PUBLIC INSTALLATION GUIDE

Native Runtime Package

Minimum public instructions for installing, updating, renewing, and safely stopping the compiled runtime used by the host application.

> **Use through the host application only**
>
> This package is not a standalone program or a general C library. Do not load the DLL or SO directly. The host application selects and verifies the correct platform file automatically.

Supported systems

System	Architecture	Python	Runtime file
Windows 10/11	x86-64	CPython 3.11, 64-bit	detection.dll
Linux	x86-64	CPython 3.10	detection.so

Quick start

1. Stop the host application before copying or replacing runtime files.
2. Copy the complete supplied native folder and preserve every filename and subfolder.
3. Confirm that the approved production model is present as models/model.engine.
4. Start the host application normally and wait for startup validation and model loading to finish.

Package and installation

Required files

Keep the supplied layout unchanged. The runtime binary, signed manifest, and XML files must belong to the same approved release.

```
native/
  windows-x86_64/detection.dll
  linux-x86_64/detection.so
  movement_tracking_manifest.json
  SystemChecksum.xml
  detection_settings.xml
```

Install

1. Stop the host application completely.
2. Copy the complete supplied native folder into the host application.
3. Preserve the supplied folder structure and filenames exactly.
4. Start the host application using its normal launcher or service.
5. Wait until startup validation and model loading report completion before using live operation.

> **Important**
>
> Do not import, rename, edit, patch, or open the DLL or SO directly. Do not combine files from different releases.

Update

1. Stop live operation and close the host application.
2. Back up the currently installed native folder.
3. Replace the DLL, SO, manifest, and XML files with the complete new release set.
4. Restart the host application and confirm that startup completes normally.

Renewal and safe operation

Renew SystemCheckscum

When an authorized replacement SystemCheckscum.xml is provided, follow these steps:

1. Stop the host application.
2. Replace only native/SystemCheckscum.xml.
3. Start the host application again and confirm that validation passes.

> **Do not edit the file**
>
> SystemCheckscum.xml must be issued by the authorized package provider. Manual changes invalidate it.

Stop safely

Use the normal service stop command or press Ctrl+C in the server console. Allow the shutdown sequence to finish before closing the console, restarting the computer, or powering it off.

Public distribution rules

Allowed	Not allowed
Approved runtime package	Private source code or generated C files
Public README and usage guide	Build tools, signing material, or test reports
Approved replacement XML files	Renamed, edited, patched, or mixed release files
Published file hashes	Debug symbols or internal documentation

> **Before distribution**
>
> Verify the supplied file hashes and provide the complete approved release as one set. Never distribute private build material with the public package.

Troubleshooting

Use the actions below before requesting a replacement package. Do not modify runtime files while troubleshooting.

Problem	Action
Native runtime validation fails	Reinstall the complete approved native release. Do not mix old and new files.
Runtime file is missing	Confirm that the supplied folder layout and filenames were preserved.
Python compatibility error	Use CPython 3.11 on Windows or CPython 3.10 on Linux.
Model cannot be loaded	Confirm that models/model.engine, the NVIDIA driver, and required GPU dependencies are installed.
Address or port is already in use	Stop the other running instance before starting the host application again.
SystemCheckscum failure	Install the current authorized SystemCheckscum.xml and restart the application.

Information to provide for support

1. Operating system and whether it is Windows or Linux.
2. Python major and minor version.
3. GPU model and installed NVIDIA driver version.
4. The exact startup error text and the time it occurred.
5. Published hashes of the installed runtime files.

> **Do not send private or unrelated data**
>
> Provide only the minimum diagnostic information listed above. Do not send credentials, private keys, source code, datasets, or unrelated system files.